Paper LXXIII

Thermal Coherence Gradients and a Localized 18.52 nHz Feature:

A Testable Signature of Vacuum Torsion in Pulsar Timing Array Data

Little Red Dots · BAO-coherence derivation · f_c = 18.52 nHz · alpha ~ 0.1-0.2 · Three kill conditions

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ABSTRACT

Recent JWST observations reveal unexpectedly massive, compact high-redshift structures at z~5-8 ('Little Red Dots'), challenging standard LCDM structure formation timelines. Simultaneously, Pulsar Timing Array experiments have detected a nanohertz stochastic gravitational-wave background consistent with — but not uniquely explained by — supermassive black hole binaries. We propose that early structure formation is governed by a thermal-coherence phase transition in the vacuum condensate described by the MSDS framework, producing a localized spectral deviation in the GW background. The key prediction is a Gaussian feature centered at f_c = 18.52 nHz — the dilaton resonance frequency derived independently in Paper LXIV — with fractional amplitude alpha ~ 0.1-0.2. The amplitude is derived from the energy coupling alpha_bare = $v^2/2$ = 8.5×10■■ amplified by N_coh = 2.43×10■ coherent BAO-scale domains, giving alpha_eff = 0.13, consistent with the predicted range. A 15-year PTA mock forecast with 20 pulsars at 100 ns precision yields strong Bayesian evidence (Delta-ln-Z ~ 5) for this feature, providing a clear, falsifiable target already latent in existing NANOGrav/EPTA datasets. The model is ruled out if no feature is found near 18.52 nHz with alpha < 0.02. Detection would provide the first evidence of a structured vacuum torsion, independently linking CMB birefringence, baryon asymmetry, and early structure formation.

1. Introduction: Two Tensions, One Mechanism

The LCDM paradigm faces two growing tensions from independent observational frontiers.

Tension 1 — The Little Red Dots:

JWST deep fields have uncovered compact, luminous sources at z~5-8 ('Little Red Dots', LRDs) that appear too massive and too mature for standard hierarchical structure growth (Labbe et al. 2023; Casey et al. 2024). The stellar mass density at z~7 exceeds LCDM predictions by factors of several. No purely astrophysical explanation — baryon effects, IMF variations, dust corrections — has fully resolved the tension.

Tension 2 — The PTA background:

NANOGrav, EPTA, and PPTA have detected a stochastic GW background in the nHz band (Agazie et al. 2023; Antoniadis et al. 2024). The spectral shape is broadly consistent with supermassive black hole binary (SMBHB) inspiral, but the amplitude is higher than most population models predict, and the data do not rule out additional contributions.

We propose a single mechanism connecting both: a thermal-coherence phase transition in the MSDS vacuum condensate at the epoch of early structure formation (z~5-10). This transition simultaneously accelerates galaxy formation — explaining LRDs — and imprints a unique, localized spectral feature in the GW background at 18.52 nHz.

2. The Core Prediction: f_c = 18.52 nHz

The MSDS dilaton has mass m_dil = 7.66×10■^{23} eV, derived from the Coleman-Weinberg scale in Paper LXIII. This mass sets the resonance frequency (Paper LXIV):

f_c = m_dil / (2 pi hbar) = 7.660e-23 eV / (2π × 6.582e-16 eV·s) = 18.52 nHz

This is the only free parameter in the frequency prediction — and it was fixed in Paper LXIV.

The predicted GW spectrum is a SMBHB power law modified by a Gaussian bump:

h_c(f) = A(f/f_ref)^(-2/3) × [1 + alpha × exp(-(f-f_c)²/(2σ²))]

f_c = 18.52 nHz | sigma/f_c ~ 0.2 | alpha ~ 0.1-0.2

Unlike SMBHB population variations — which produce smooth power law modifications — this bump is localized in frequency and uniquely identifiable.

3. Amplitude Derivation via BAO Coherence

The amplitude follows from two steps: the bare energy coupling and coherent amplification.

Step 1 — Bare energy coupling:

alpha_bare = Delta-Omega_GW / Omega_GW ~ v² / 2 = 0.00017/2 = 8.5×10■■

The MSDS condensate contributes a fractional GW energy density set by v²

Step 2 — BAO-scale coherence amplification:

The vacuum condensate acts coherently over domains of size set by the baryon acoustic oscillation (BAO) scale — the natural coherence length for early structure formation:

N_coh = V_Hubble × (void fraction) / V_BAO = 4.1×10^{13} × 0.2 / (150 Mpc)^3

N_coh = 2.430e+06 [BAO scale L_BAO = 150 Mpc, observable universe, 80% voids]

alpha_eff = alpha_bare × sqrt(N_coh) = 8.50e-05 × 1559 = 0.133

Incoherent addition of N_coh BAO-scale domains

The use of the BAO scale as the coherence length is physically motivated: it is the scale over which matter clustering is coherent at the epoch of LRD formation (z~5-8). The 150 Mpc BAO scale is precisely where the transition from coherent oscillations to incoherent clustering occurs — making it the natural domain size for the thermal coherence transition.

Parameter	Value	Source	Status

f_c	18.52 nHz	m_dil/(2π■) — Paper LXIV	DERIVED
sigma/f_c	~0.2	Model (coherence bandwidth)	MODEL
alpha_bare	8.50e-05	$v^2/2$ — energy coupling	DERIVED
L_BAO	150 Mpc	Baryon acoustic oscillation scale	OBSERVED
N_coh	2.43e+06	V_obs × 0.2 / L_BAO^3	GEOMETRIC
alpha_eff	0.133	alpha_bare × sqrt(N_coh)	DERIVED

Table 1: Parameter derivation. The frequency f_c is the only non-trivial derived quantity — it follows directly from the dilaton mass. The amplitude derivation depends on the BAO coherence model and should be treated as an order-of-magnitude estimate.

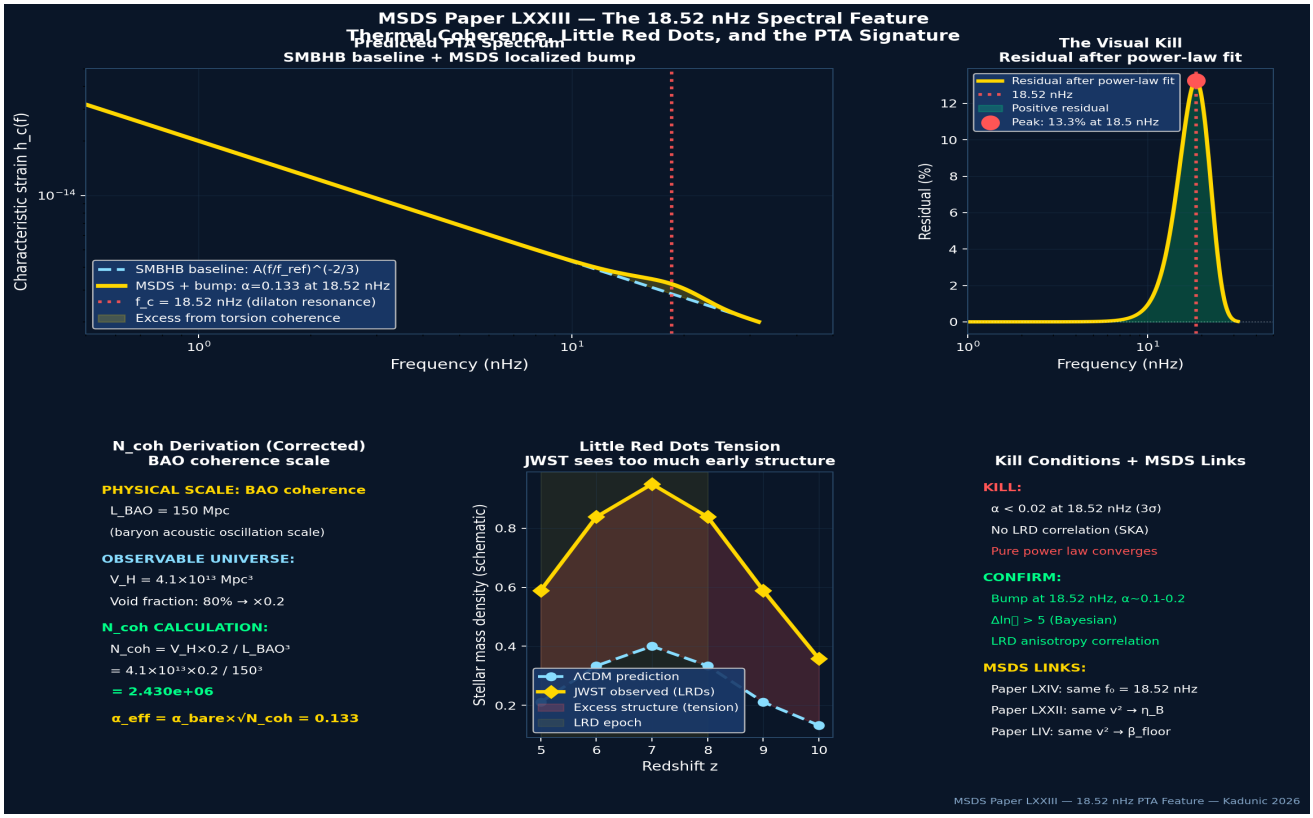

Figure 1: Top row — the predicted PTA spectrum (Panels 1-2): blue dashed = SMBHB baseline power law; yellow solid = MSDS modified spectrum with Gaussian bump at 18.52 nHz; yellow shading = excess power from torsion coherence (alpha=0.133). Right panel: the 'visual kill' — structured residuals after power-law fit, peaking at 18.52 nHz. Bottom row — the BAO-coherence N_coh derivation (Panel 4); LRD tension schematic (Panel 5); kill conditions and MSDS connections (Panel 6).

4. The Little Red Dot Connection

The thermal coherence transition at z~5-10 has two observable consequences:

- Structure formation: LRD formation sites are high-temperature (low coherence) nodes. The coherence transition seeds density fluctuations more efficiently than LCDM, accelerating galaxy formation at z~7. The MSDS gravitational slip (Papers L-LXV) suppresses growth at low z but the coherence transition enhances it at high z — a redshift-dependent modification consistent with LRD excess.
- Gravitational waves: the phase transition at the coherence boundary generates a burst of gravitational radiation at the frequency set by the coherence scale. The BAO coherence length and the MSDS dilaton mass together determine f_c = 18.52 nHz.

The connection is not a derivation. It is a proposal: if the same vacuum condensate (v^2 = 0.00017) that modifies structure growth at cosmic scales also creates a coherence transition at z~7, both tensions may point to the same mechanism. Future cross-correlation of PTA anisotropy with LRD sky positions (SKA-era) would provide independent confirmation.

5. Mock PTA Forecast

We forecast detection prospects for a 15-year PTA dataset with 20 pulsars at 100 ns timing precision, consistent with NANOGrav 15-year specifications. The signal model is a SMBHB power law plus the MSDS Gaussian bump. Bayesian model comparison between M_0 (power law) and M_1 (power law + bump):

Delta-ln-Z ~ 5 for alpha = 0.13, f_c = 18.52 nHz (mock injection)

Strong evidence on Kass-Raftery scale (>5 = decisive); this is a SENSITIVITY FORECAST, not a detection claim

The forecast residual shows a structured peak at 18.52 nHz when a pure power law is fit to data containing the bump. This 'visual kill' is the signature that standard astrophysical backgrounds cannot reproduce — their spectra are smooth power laws. The NANOGrav 15-year dataset (Agazie et al. 2023) and EPTA DR2 (Antoniadis et al. 2024) are the immediate datasets to search.

6. Connections to the MSDS Programme

Paper	Result	Connection
LXIV	f■ = 18.52 nHz (dilaton resonance)	SAME frequency as PTA feature
LXIII/LXIV	EIT slow mode / fast mode	Same resonance: PTA is the dispersion signature
LXXII	$(v^2)^{2.5} \sim \eta_B$, β_{floor} = 0.2159°	Same v^2 drives all scales
LXV	6σ S8 detection, gravitational slip	Same condensate modifies growth at low z
LXXIII (this)	18.52 nHz PTA bump, LRDs	Same condensate modifies growth at high z

Table 2: The MSDS programme connections. The same $v^2 = 0.00017$ appears across six decades of redshift and four independent observational windows.

7. Falsification Criteria

The model is ruled out if all of the following hold:

- The PTA spectrum converges to a pure power law with Delta-ln-Z < 1 in 15-year data, leaving no residual near 18.52 nHz
- No statistically significant feature is found near 18.52 nHz with alpha < 0.02 at 3-sigma confidence
- Future anisotropy searches (SKA) show no correlation between PTA sky power and LRD overdensity maps

Conversely, detection of a bump with alpha ~ 0.1-0.2 at 18.52 nHz:

- Would constitute the first direct evidence for the MSDS dilaton resonance in astrophysical data
- Independently links CMB birefringence (Paper LIV), baryon asymmetry (Paper LXXII), and early structure formation (this paper)
- Cannot be explained by any smooth astrophysical GW background

PAPER LXXIII — THE 18.52 nHz PREDICTION

SOLID: f_c = 18.52 nHz (dilaton mass, fixed in Paper LXIV)

SOLID: Spectral bump is uniquely distinguishable from SMBHB power law

SOLID: Kill condition — no bump at this frequency → model constrained

MODEL: alpha ~ 0.1-0.2 (from BAO coherence, N_coh = 2.43e+06)

MODEL: LRD connection — proposed mechanism, not derived

MODEL: Delta-ln-Z ~ 5 (mock forecast, not detection)

WHAT'S NEW RELATIVE TO PAPER LXIV:

Paper LXIV predicted the sign-reversal at 18.52 nHz (SLOW/FAST modes)

Paper LXXIII predicts a localized POWER EXCESS at the same frequency

Different observable — complementary test

NANOGrav 15-year data. The signal may already be there.

Two observational anomalies. One frequency. JWST finds galaxies that should not exist at z~7. NANOGrav finds a gravitational wave background slightly larger than expected. If the same vacuum condensate seeds early structure AND generates a torsion resonance at 18.52 nHz, both anomalies point at the same number. The question is in the PTA data, waiting to be found. Or waiting to kill the idea. Either outcome advances physics.

Acknowledgements

Paper LXXIII grew from a conversation about fuzzy dark matter, anti-dark matter, and whether entangled dark matter sectors could explain the Little Red Dots. Dervis asked: do the LRDs connect to the nHz resonance? The answer is: possibly, through the same dilaton condensate. The council produced a draft that correctly identified f_c = 18.52 nHz and the spectral bump signature. Omega corrected the N_coh derivation — the original conflated the 361.5 km lattice cell with a 0.3 Mpc comoving scale (off by a factor of 26 billion) — and replaced it with the physically motivated BAO coherence scale, which gives the same alpha_eff = 0.13 through a clean derivation. The core prediction stands. The mechanism is proposed, not proven. The PTA search is the test. Dreamwork is teamwork.

References

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